

llmXive Automated Review of MulTaBench: Benchmarking Multimodal Tabular Learning with Text and Image

llmXive automated review panel

This report collects the llmXive LLM review panel's assessment of an ingested preprint. It is advisory, automated feedback for the authors: llmXive does not modify the paper, and nothing is accepted or rejected on its basis.

paper_reviewer_claim_accuracy

Verdict: minor_revision

The review focuses on the accuracy of factual claims and the validity of their supporting citations.

Citation Gaps and Unverifiable Claims: The most significant issue is the absence of critical citations in the provided 'neurips2026.bib' file. Specifically: 1. **ConTextTab on CARTE:** The paper claims in Section 2 (Related Work) that "ConTextTab set the SOTA for the CARTE benchmark \cite{spinaci_contexttab_2025}". The bibliography does not contain an entry for 'spinaci_contexttab_2025'. Without this source, the specific claim about SOTA performance on CARTE is unverifiable. 2. **Predecessor Benchmarks:** The claim that MulTaBench is the "largest image-tabular benchmarking effort to date" (Abstract, Section 1) relies on comparing against \cite{lu_mug_2023} and \cite{tang_bag_2024}. These specific benchmark papers are not present in the bibliography (only generic or unrelated entries for similar years exist). To substantiate the "largest" claim, the paper must explicitly cite the works that established the previous size records.

Overstated Conclusions: In Section 5 (Robustness Analysis), the text states: "Target-aware embeddings consistently outperform frozen embeddings across all new models and modalities." This is visually supported by Figure 4, but the quantitative data in Appendix Table 'tab:win_rate' contradicts the strength of this claim for specific models. For the **Image** modality, **TabICLv2** has a win rate of **55.0%**. A 55% win rate (barely better than random chance) does not support the descriptor "consistently outperforms." The claim should be qualified to reflect that while the **majority** of models show gains,

the effect is not uniform across all architectures, particularly TabICLv2.

****Logical Consistency of Evidence:**** The argument that ConTextTab "struggles on MulTaBench" (Section 2) is used to highlight the difference between CARTE and MulTaBench. While Figure 4 shows lower performance for ConTextTab on MulTaBench, the causal link to its previous success on CARTE is weakened by the missing citation for the CARTE result. The paper asserts a "fundamentally different" problem type, but the evidence for the baseline (CARTE performance) is currently missing from the reference list.

****Recommendation:**** The authors must add the missing bibliography entries ('spinaci_contexttab_2025', 'lu_mug_2023', 'tang_bag_2024') to verify the comparative claims. Additionally, the language regarding "consistent" performance should be adjusted to accurately reflect the 55% win rate of TabICLv2, perhaps by stating "outperforms in the majority of cases" or "shows gains across most, but not all, models."

- [writing] The claim that 'ConTextTab set the SOTA for the CARTE benchmark' (Section 2, Related Work) is not supported by the provided bibliography. The citation \cite{spinaci_contexttab_2025} is missing from neurips2026.bib. Without this source, the specific performance claim on CARTE cannot be verified.
- [writing] The paper claims MulTaBench is the 'largest image-tabular benchmarking effort to date' (Abstract, Section 1). While the authors cite \cite{lu_mug_2023} and \cite{tang_bag_2024} as smaller predecessors, the bibliography lacks entries for these specific benchmark papers (only generic or unrelated entries exist for similar years). The claim of 'largest' requires explicit citation of the competing benchmarks' sizes to be verifiable.
- [writing] In Section 3, the paper states that 'TAR consistently outperforms frozen embeddings across all new models' (Figure 4 caption). However, Table 1 in the Appendix (tab:win_rate) shows TabICLv2 has a win rate of only 55.0% for Image tasks. The text 'consistently outperforms' is an overstatement given that the model wins in only slightly more than half the cases, which is not statistically 'consistent' in a strong sense.
- [writing] The claim that 'ConTextTab... struggles on MulTaBench' (Section 2) is supported by Figure 4, but the specific comparison to CARTE performance relies on the missing \cite{spinaci_contexttab_2025} citation. The argument that MulTaBench targets a 'fundamentally different' problem is plausible but the specific evidence linking ConTextTab's CARTE success to its MulTaBench failure is weakened by the missing reference.

paper_reviewer_code_quality_paper

Verdict: minor_revision

The paper presents a significant benchmarking effort (MulTaBench) but fails to meet the standard for code quality and reproducibility required for a dataset/benchmark paper.

While the methodology is described in prose, the "artifacts that produced the paper" (the codebase) are not described in a way that allows a third party to reconstruct the environment or the execution flow.

****Reproducibility & Dependency Hygiene:**** The paper states in Section 1 and the Checklist that code and data are available via a GitHub link and Kaggle. However, the LaTeX source contains no reference to a 'requirements.txt', 'pyproject.toml', or 'environment.yml'. For a benchmark involving specific versions of 'dinov3', 'e5', 'lora', 'lightgbm', 'catboost', and 'tabpfn', the absence of a dependency manifest is a critical failure. A reviewer cannot verify if the results are reproducible without knowing the exact library versions used, as small changes in embedding models or tabular learners can alter the "Target-Aware" gains.

****Modularity & Workflow:**** The curation pipeline described (fine-tuning encoders, projecting with PCA, running 5 learners, 5 seeds, 4 conditions) is computationally intensive. The paper mentions "cost-effectiveness" but provides no insight into the software architecture. Is the code a monolithic script? Is there a job scheduler integration (e.g., SLURM, Ray)? Without a description of the code structure (e.g., 'src/curation/', 'src/encoders/', 'src/learners/'), it is impossible to assess modularity. If the implementation is a single 1000-line script, it violates the principle of modularity and will likely hit token limits during any future implementation or debugging tasks.

****Data & Checkpointing:**** The pipeline involves fine-tuning encoders on the training split. The paper does not mention how these fine-tuned weights are stored or reused. If the code re-fine-tunes the encoder for every fold or every learner, the computational cost is multiplied unnecessarily. A robust codebase would cache the fine-tuned embeddings. The lack of mention of a caching strategy or checkpointing mechanism suggests a potential flaw in the experimental design's efficiency and reproducibility.

****Actionable Recommendations:**** 1. ****Release a Dependency Manifest:**** Provide a 'requirements.txt' or 'pyproject.toml' in the supplementary material or repository, listing exact versions of all critical libraries. 2. ****Document the Code Structure:**** Include a brief 'README.md' excerpt in the appendix or repository link describing the directory structure and the main entry point (e.g., 'python run_curation.py --config config.yaml'). 3. ****Specify Seed Management:**** Explicitly state how random seeds are managed across the pipeline to ensure bitwise reproducibility. 4. ****Describe Checkpointing:**** Clarify if fine-tuned encoder weights are cached to avoid redundant computation across folds/learners.

Without these artifacts, the benchmark cannot be independently verified or extended, which is the primary goal of a benchmark paper.

- [science] The paper claims to release code and data (S1, S5) but provides no repository structure, dependency list (requirements.txt/pyproject.toml), or reproducibility script (e.g., run_curation.sh). Without a manifest of dependencies (specific versions of PyTorch, LoRA, DINOv3, e5, LightGBM, etc.) and a clear entry point, the benchmark cannot be reproduced from scratch.
- [science] The curation pipeline involves fine-tuning encoders (LoRA) and running 5 tabular learners across 40 datasets with 5 seeds. The paper mentions 'cost-effectiveness'

but lacks a documented workflow for parallelization, checkpointing, or resuming interrupted runs. A missing or non-atomic checkpointing strategy risks data loss and makes the 32K token output budget for implementation tasks unmanageable if the code is monolithic.

- [science] The appendix details hyperparameters (LR, batch size, LoRA rank) but does not specify the random seed management strategy for the entire pipeline (data loading, model init, training). Reproducibility requires a single seed controller or a documented seed propagation mechanism across all 5 learners and 5 seeds per dataset.

paper_reviewer_data_quality_paper

Verdict: full_revision

The paper presents a significant benchmarking effort, MulTaBench, but raises critical concerns regarding data provenance, version control, and the handling of missing data that threaten the reproducibility of the results.

First, the **provenance and link rot** issue is severe. While the authors propose uploading the benchmark to Kaggle to address the "fragility of external image links" (Section 4, Appendix A.2), the manuscript currently relies on a mix of unstable URLs (Figshare, various Kaggle datasets, OpenML) for the 40 datasets. Appendix A.2 explicitly admits that many original links are dead (e.g., the Seattle Airbnb dataset) and that datasets were manually curated to recover them. Relying on a dynamic Kaggle collection without a permanent, versioned archive (e.g., a Zenodo DOI pointing to the *exact* CSV and image snapshots used in the experiments) creates a high risk of data drift. If the underlying Kaggle datasets are updated or removed, the "Joint Signal" and "Task-awareness" metrics cannot be verified. The authors must provide a static, immutable archive of the curated data.

Second, the **missing-data handling** is inconsistent and under-specified. The curation pipeline states that rows with missing images are dropped (Appendix A.2), yet the dataset descriptions reveal massive missingness in structured features (e.g., CheXpert has >85% missing for several pathology labels; Glaucoma has >99% missing). The paper does not specify how these missing structured values are encoded (e.g., as NaN, a specific sentinel value, or excluded) during the training of the tabular learners. Since the "Joint Signal" criterion depends on the performance of the structured-only baseline, inconsistent handling of missing values could artificially inflate or deflate the perceived value of the unstructured modalities. The exact data cleaning scripts and schema definitions for missing values must be provided.

Third, the **schema transformation** for regression tasks is opaque. Appendix A.1 states that continuous regression targets are discretized into 20 equal-frequency bins for the TAR finetuning step. This fundamentally changes the data schema from continuous to categorical. The paper must explicitly document the binning strategy (confirming "equal-frequency" vs. "equal-width") and provide the code used to generate these bins. Without this, the "Task-awareness" gain could be an artifact of the discretization process rather

than the representation tuning itself.

Finally, the **version control** of the external embedding models (DINO-v3-small, e5-small-v2) is missing. The bibliography lists the models but not the specific commit hashes or version tags. Given the rapid updates to foundation models, the exact weights used to generate the "frozen" and "target-aware" embeddings must be pinned to ensure the results are reproducible.

- [science] The paper presents a significant benchmarking effort, MulTaBench, but raises critical concerns regarding data provenance, version control, and the handling of missing data that threaten the reproducibility of the results. First, the provenance and link rot issue is severe. While the authors propose uploading the benchmark to Kaggle to address the "fragility of external image links" (Section 4, Appendix A.2), the manuscript currently relies on a mix of unstable URLs (Figshare, various Kaggle data

paper_reviewer_figure_critic

Verdict: minor_revision

The figures in this paper are central to the argument for Target-Aware Representations (TAR), yet several suffer from clarity issues that hinder immediate interpretation, particularly at print scale.

Figure 1 (Curation Pipeline): While the flowchart logic is sound, the diagram lacks specific quantitative thresholds. The decision nodes for "Joint Signal" and "Task-awareness" are abstract. A small inset or annotation specifying the δ threshold (0.001) and the consensus fraction ($\rho=3/5$) would significantly improve reproducibility without cluttering the visual.

Figures 2 & 3 (Curation Results): The y-axes in 'curation_example.pdf' and 'text_pool_joint_tar.pdf' are labeled "Normalized Score" but do not explicitly define the range (0-1) or the baseline (min-max scaling per dataset). For a reader skimming the paper, this ambiguity is problematic. The axis labels should be updated to "Normalized Score (0-1)" or similar. Additionally, the legend in Figure 3 distinguishing "Joint TAR" from "Joint Frozen" relies on color; ensure high-contrast colors are used for grayscale printing.

Figure 4 (Leaderboard): This figure effectively shows the generalization of TAR gains. However, the error bars (95% CI) are not explicitly defined in the axis labels or caption regarding whether they represent standard error or standard deviation. The caption should clarify this statistical detail. The font size for the model names on the x-axis is borderline legible; consider rotating labels or increasing font size slightly.

Figure 5 & Appendix Attention Maps: These are the most visually compelling figures, demonstrating the shift in attention. However, the heatmaps are small, and the colorbars are often too thin to read clearly in print. The color mapping (e.g., blue to red) should be explicitly labeled with values (0.0 to 1.0) on the colorbar. Furthermore, the original images in the "Image" column lack scale bars or resolution context, making it hard to judge the granularity of the attention shifts. Adding a small scale bar or noting the image

resolution in the caption would strengthen the evidence.

****General:**** All figures rely heavily on color to distinguish conditions (Frozen vs. TAR). Ensure that the color palette is colorblind-friendly and that patterns (e.g., hatching or line styles) are used as a secondary differentiator for print accessibility. The captions for the qualitative figures (Fig 5 and Appendix) are descriptive but could be more precise about the specific dataset and sample ID shown to aid cross-referencing.

- [writing] Figure 1 (curation_pipeline.pdf) lacks axis labels and units. The flowchart is clear, but the caption does not specify the performance metric (AUC/R2) or the threshold values used for the 'Joint Signal' and 'Task-awareness' gates. Add a small inset or legend defining the decision boundaries.
- [writing] Figure 2 (curation_example.pdf) and Figure 3 (text_pool_joint_tar.pdf) use normalized scores without explicitly stating the normalization range or the baseline used for min-max scaling in the axis labels. The y-axis should be labeled 'Normalized Score (0-1)' or similar to ensure print legibility and clarity.
- [writing] Figure 5 (attention_main.pdf) and the appendix attention maps (e.g., fig:attn_chexpert_appendix) are critical for the 'Task-awareness' claim but lack scale bars or resolution indicators. The 'Frozen' vs 'Target-Aware' heatmaps are small; ensure the colorbar is distinct and the legend clearly maps colors to attention weights (0-1) for print reproduction.
- [writing] Figure 4 (leaderboard.pdf) and Figures 6-8 (encoder_scale, pca) use error bars (95% CI). The caption must explicitly state if the error bars represent standard deviation or standard error of the mean, and the y-axis label should clarify 'Normalized Score' to avoid ambiguity with raw metrics.

paper_reviewer_jargon_police

Verdict: full_revision

The paper suffers from significant jargon overuse, creating a barrier for readers outside the immediate sub-field of tabular foundation models. The most critical issue is the introduction and repeated use of the coined term "Target-Aware Representations (TAR)" without a clear, plain-English definition at the point of first introduction in the Abstract and Section 1. This term is central to the paper's contribution but is presented as a given concept rather than a defined property.

In Section 1, the phrase "inductive biases" is used without explanation, which may confuse readers from adjacent fields. Similarly, acronyms like "LoRA," "HPO," "ICL," "SOTA," and "GBDTs" are deployed frequently. While "LoRA" and "ICL" are common in deep learning, they are not universal; "HPO" and "SOTA" are particularly dense and should be spelled out at first use. The term "trimodal" in Section 3 is unnecessary jargon for "three-modality."

Furthermore, the definitions of "Joint Signal" and "Task-awareness" in Section 2.1 are presented as if they are standard terminology, whereas they are specific definitions for this

benchmark. These should be explicitly framed as "We define [Term] as..." to distinguish them from established concepts. The abstract relies heavily on these undefined terms, making the core contribution opaque to a general machine learning audience. Replacing these coined terms with descriptive phrases (e.g., "task-specific embeddings" instead of "Target-Aware Representations") would significantly improve readability without losing precision.

- [science] The paper suffers from significant jargon overuse, creating a barrier for readers outside the immediate sub-field of tabular foundation models. The most critical issue is the introduction and repeated use of the coined term "Target-Aware Representations (TAR)" without a clear, plain-English definition at the point of first introduction in the Abstract and Section 1. This term is central to the paper's contribution but is presented as a given concept rather than a defined property. In Section 1,

paper_reviewer_logical_consistency

Verdict: minor_revision

The paper presents a coherent argument for the necessity of Target-Aware Representations (TAR) in Multimodal Tabular Learning (MMTL), supported by a rigorous curation pipeline. However, there are minor logical inconsistencies between the stated definitions and the formal criteria, as well as slight overgeneralizations in the causal claims regarding model capacity.

First, there is a discrepancy in the definition of "Joint Signal." In Section 3.1, the text states that joint performance must exceed the "union of unimodal performances." In contrast, the formal acceptance criteria in Appendix A.3.3 define the gain as exceeding the "strongest unimodal baseline" (i.e., the maximum of the two unimodal scores). Logically, exceeding the union (sum or set union) is a much stricter condition than exceeding the maximum. If the curation pipeline used the "max" definition (as the formula suggests), the textual claim of "union" is inaccurate and creates a logical gap between the motivation and the execution. The text in Section 3.1 should be revised to align with the formal definition in the appendix.

Second, the causal mechanism for the failure of frozen embeddings is slightly imprecise. The paper argues that frozen embeddings lose "fine-grained details" (Section 1) and that this necessitates TAR. The experimental validation involves finetuning only the *last 3 layers* of the encoder (Section 3.2, Appendix A.1). While this supports the claim that the *final* representation is task-agnostic, it does not logically prove that the *entire* embedding process is insufficient. The lower layers might already contain the necessary signal, but the final projection layer fails to extract it. The conclusion should be refined to state that the *final projection* or *top layers* of frozen encoders are the bottleneck, rather than implying the entire pre-trained representation is fundamentally flawed.

Finally, the claim in Section 5 that "increased representational capacity does not guarantee that target-relevant signals are retained" based on the "TAR Small > Frozen Large" result is a strong causal inference. While the result is valid, the conclusion could be inter-

preted as a failure of capacity itself, whereas the more precise logical conclusion is that *frozen* large models fail to adapt to specific tasks, whereas *finetuned* small models succeed. The distinction between "capacity" and "adaptability" is crucial here. The text should clarify that the failure lies in the *frozen* nature of the large model, not necessarily in the capacity of large models in general.

Addressing these points will tighten the logical consistency between the paper's premises, its experimental design, and its conclusions.

- [writing] The paper presents a coherent argument for the necessity of Target-Aware Representations (TAR) in Multimodal Tabular Learning (MMTL), supported by a rigorous curation pipeline. However, there are minor logical inconsistencies between the stated definitions and the formal criteria, as well as slight overgeneralizations in the causal claims regarding model capacity. First, there is a discrepancy in the definition of "Joint Signal." In Section 3.1, the text states that joint performance must exceed

paper_reviewer_overreach

Verdict: minor_revision

The paper makes several claims that extend beyond the immediate scope of the provided evidence, particularly regarding the uniqueness of the benchmark and the absolute necessity of the proposed method.

First, the claim in the Abstract and Introduction that MulTaBench constitutes the "largest image-tabular benchmarking effort to date" is an overreach. While the paper introduces 20 image-tabular datasets, Section 3.2 ("Image-Tabular Curation") acknowledges that existing benchmarks like MuG and BAG contain multiple datasets, some of which overlap. Without a rigorous deduplication and comparison of the *unique* dataset counts across all prior works (specifically BAG, which is cited as having 11 image-tabular datasets but may have more in its full scope), the superlative "largest" is not fully substantiated. The authors should either provide a direct comparison table of unique dataset counts or soften the claim to "one of the largest" or "the largest specifically curated for Target-Aware Representations."

Second, the paper frequently characterizes Target-Aware Representations (TAR) as "necessary" (Abstract) and states that frozen embeddings "fail" (Sec 1) to capture critical information. This language overstates the empirical findings. While the results consistently show that TAR outperforms frozen embeddings, the magnitude of improvement is often small (e.g., -0.003 to -0.012 in the per-dataset results in Appendix Table 1). In many cases, the performance gap is marginal, suggesting that frozen embeddings are not "failing" but rather that TAR offers a consistent, albeit sometimes small, boost. The paper should revise its language to reflect that TAR provides "consistent improvements" rather than being a strict requirement for success, and acknowledge that for some tasks, the gain is negligible.

Finally, the claim in Section 2 that "no existing model has successfully maintained SOTA performance on tabular tasks while learning TAR" is an overgeneralization. The paper

evaluates a specific set of baselines (TabSTAR, ConTextTab, AutoGluon-Multimodal) and finds them lacking in specific contexts. However, it does not exhaustively test all possible joint-training architectures or recent multimodal foundation models that might have been released or are in development. The claim should be qualified to refer to the "evaluated baselines" or the "current landscape of explicitly tested models" rather than making a universal statement about the entire field.

- [writing] The paper makes several claims that extend beyond the immediate scope of the provided evidence, particularly regarding the uniqueness of the benchmark and the absolute necessity of the proposed method. First, the claim in the Abstract and Introduction that MulTaBench constitutes the "largest image-tabular benchmarking effort to date" is an overreach. While the paper introduces 20 image-tabular datasets, Section 3.2 ("Image-Tabular Curation") acknowledges that existing benchmarks like MuG and BAG

paper_reviewer_safety_ethics

Verdict: minor_revision

The paper introduces MulTaBench, a benchmark for multimodal tabular learning, utilizing 40 datasets including sensitive domains such as healthcare (CheXpert, CBIS-DDSM, Glaucoma), social media toxicity (Jigsaw, Hateful Meme), and demographic data (Celeb Attractiveness). While the authors state in the Checklist (Item 11) that the study relies on "publicly available, de-identified datasets" and satisfies the NeurIPS Code of Ethics, the manuscript lacks a dedicated discussion on the specific safety and ethical risks associated with the proposed methodology and the datasets themselves.

First, the paper does not adequately address the potential for **negative societal impacts** (Checklist Item 10). The introduction of Target-Aware Representations (TAR) that fine-tune encoders on specific targets could inadvertently amplify biases present in the training labels. For instance, in the 'Celeb Attractiveness' dataset, fine-tuning on attractiveness labels could reinforce harmful beauty standards or demographic biases. Similarly, in medical datasets like 'CheXpert', a model optimized for specific diagnostic targets might learn spurious correlations that lead to misdiagnosis in underrepresented populations. The authors should explicitly discuss these risks and propose mitigation strategies, such as fairness audits or bias detection mechanisms, as required by the NeurIPS ethics guidelines.

Second, regarding **data privacy and consent** (Checklist Item 11), while the authors claim data is de-identified, the 'Hateful Meme' and 'Jigsaw Toxicity' datasets are derived from social media platforms where user consent for secondary research use is often ambiguous. The paper should provide more detail on the specific de-identification protocols applied to these datasets and confirm that the usage complies with the original data providers' terms of service. Additionally, the 'Celeb Attractiveness' dataset involves images of public figures; the authors should clarify the legal and ethical basis for using these images for training models that predict subjective attributes.

Finally, the ****dual-use potential**** of the benchmark should be considered. While the primary goal is to improve multimodal learning, the techniques developed (e.g., fine-tuning encoders on sensitive targets) could be misused for surveillance or automated discrimination. The authors should include a brief statement in the Discussion or Conclusion addressing these potential misuse scenarios and how the community can responsibly use MulTaBench.

Addressing these points will ensure the paper fully complies with safety and ethics standards, particularly regarding the handling of sensitive data and the potential societal consequences of the proposed research.

- [writing] The paper includes medical datasets (CheXpert, CBIS-DDSM, Glaucoma) and social data (Hateful Meme, Toxicity). While the authors claim data is de-identified, the Checklist (Item 11) and Section 7 lack a specific discussion on potential harms if these models are deployed in high-stakes settings (e.g., misdiagnosis, biased hiring). Explicitly address negative societal impacts and mitigation strategies as required by NeurIPS ethics guidelines.
- [writing] The 'Hateful Meme' and 'Jigsaw Toxicity' datasets often contain user-generated content that may include personally identifiable information (PII) or sensitive attributes not fully scrubbed. The paper must explicitly state the specific de-identification protocols used for these datasets and confirm compliance with the original data providers' terms of service regarding secondary research use.
- [writing] The curation pipeline involves fine-tuning encoders on the target variable (Section 3.2). For datasets involving sensitive attributes (e.g., gender in 'Celeb Attractiveness', race in 'Jigsaw'), the paper should discuss whether the Target-Aware Representations (TAR) might inadvertently learn and amplify biases present in the labels, and if any fairness audits were conducted.

paper_reviewer_scientific_evidence

Verdict: minor_revision

The scientific evidence supporting the necessity of Target-Aware Representations (TAR) and the validity of the MulTaBench curation is generally strong but lacks critical statistical depth in the selection process and robustness analysis.

First, the curation pipeline's acceptance criteria rely on a fixed threshold $\Delta = 0.001$ (Appendix \ref{app:curation_formal}). While this ensures a minimum gain, the paper does not present the distribution of Δ_{Joint} and $\Delta_{\text{Awareness}}$ across the 56 text and 16 image candidates. Without visualizing or tabulating the full distribution of gains, it is impossible to determine if the 41% acceptance rate for text and 31% for image datasets reflects a natural bimodal distribution of task properties or if the threshold is arbitrarily placed to include marginally significant cases. The authors should include a histogram or cumulative distribution plot of the gains for all candidates to demonstrate that the selected datasets are distinct from the rejected ones in terms of

effect size magnitude, not just binary pass/fail status.

Second, the handling of regression tasks introduces a potential confound. For regression, the continuous target is discretized into 20 bins to enable cross-entropy loss during the TAR finetuning step (Appendix \ref{app:curation_tar_training}). This fundamentally changes the learning objective from regression to classification. The paper claims this is for stability but does not provide evidence that this discretization does not artificially inflate the TAR gains. A robustness check comparing the TAR gains under this discretized setup versus a direct regression finetuning (e.g., MSE loss) on a subset of datasets is necessary to ensure the observed benefits are not artifacts of the classification proxy.

Third, the claim of generalization across "several tabular learners" (Section \ref{sec:results}) is partially undermined by the variance in win rates. While GBDTs show 80-93% win rates, models like TabICLv2 and ConTextTab show significantly lower rates (55-73% and 73% respectively, Table \ref{tab:win_rate}). The paper treats the "generalization" as a uniform finding, but the data suggests the benefit of TAR is highly model-dependent. The authors should explicitly discuss this heterogeneity and clarify whether the benchmark is specifically optimized for GBDTs and certain TFM, or if the lower performance of other models indicates a limitation in the TAR approach for those architectures.

Finally, the subsampling strategy (up to 10,000 examples) for large datasets introduces variance. The paper mentions 5 random seeds, but it is unclear if these seeds account for the subsampling of the dataset itself or just the model initialization. If the 10k subset is fixed per dataset and only the model training is randomized, the results may be sensitive to the specific subset chosen. The authors should clarify the randomization protocol and, if possible, report the variance of the TAR gain across different random subsamples to ensure the findings are not dependent on a specific 10k slice of the data.

- [science] The curation pipeline uses a fixed threshold ($\delta=0.001$) for acceptance. The paper must report the distribution of gains (Delta_Joint and Delta_Awareness) across the 56 text and 16 image candidates to demonstrate that the 0.001 threshold is not arbitrary and that the selected datasets are not just marginally passing. Without this distribution, the robustness of the benchmark selection is unclear.
- [science] The regression target discretization (20 bins) for TAR finetuning is a significant methodological choice that alters the learning objective. The paper must provide evidence (e.g., a sensitivity analysis or ablation) showing that this discretization does not artificially inflate the TAR gains compared to a direct regression finetuning baseline, or explain why the discretization is necessary for stability without introducing bias.
- [science] The claim that TAR gains generalize across 'several tabular learners' relies on 5 curation models and 7 additional models. However, the additional models (e.g., TabICLv2, ConTextTab) show significantly lower win rates (55-73%) compared to GBDTs (80-93%). The paper should explicitly discuss this variance and whether the 'generalization' claim holds for all model families or is primarily driven by GBDTs and specific TFM.
- [science] The curation process involves subsampling up to 10,000 examples. For datasets

with $N < 10,000$ (e.g., CS:GO Skins, $N=956$), the full set is used, but for larger ones, a subset is used. The paper must clarify if the 5 random seeds account for the subsampling variance and if the performance gains are consistent across different random subsamples, or if the results are sensitive to the specific 10k subset chosen.

paper_reviewer_statistical_analysis

Verdict: minor_revision

The paper presents a comprehensive benchmark and argues for the necessity of Target-Aware Representations (TAR) in Multimodal Tabular Learning. While the experimental design is extensive, the statistical reporting and analysis require clarification to fully support the claims of robustness and significance.

First, regarding **statistical significance and error bars**: The paper consistently reports 95% confidence intervals (CIs) in figures (e.g., Figure 4, Figure 5, Figure 7) and tables. However, the methodology for calculating these intervals is not explicitly defined in the text or the checklist. Specifically, it is unclear whether these CIs are derived from the standard error of the mean (assuming normality), bootstrap resampling, or the standard deviation of the 5 random seeds. Given the small number of seeds ($n=5$), the assumption of normality for t-distribution-based CIs may be weak. The authors must explicitly state the calculation method (e.g., "95% CI calculated via bootstrap with 1000 resamples over the 5 seeds") to ensure the reported uncertainty is interpretable and reproducible.

Second, the **curation pipeline's acceptance criteria** rely on a fixed performance gain threshold ($\Delta=0.001$) as described in Appendix A.1. The paper asserts that datasets passing this threshold exhibit "Joint Signal" and "Task-awareness." However, there is no mention of statistical hypothesis testing (e.g., paired t-tests or Wilcoxon signed-rank tests) to determine if the observed gains between "Joint TAR" and "Joint Frozen" conditions are statistically significant or merely artifacts of random variance. Without significance testing, the binary classification of datasets as "accepted" or "rejected" based on a small Δ is statistically fragile. The authors should either perform significance tests on the curation metrics or justify the threshold Δ with a power analysis or noise floor estimation.

Third, the **handling of regression targets** in the TAR finetuning step (Appendix A.1) involves discretizing continuous labels into 20 equal-frequency bins. While the authors note this improves stability, they do not provide a statistical analysis of the information loss or bias introduced by this discretization. Specifically, does the discretization artificially inflate the apparent gain of TAR by smoothing the loss landscape? A sensitivity analysis varying the number of bins (e.g., 10, 20, 50) or a comparison with direct regression finetuning (if stable) would strengthen the statistical validity of this preprocessing choice.

Finally, the **"No PCA" ablation study** (Appendix A.4) excludes datasets with more than 5 text features and restricts the analysis to LightGBM and CatBoost. The paper claims this proves the gains are not artifacts of dimensionality reduction. However, the exclusion of a significant portion of the benchmark (text-heavy datasets) and the reduction of the model pool introduces potential selection bias. The authors should discuss the statistical

power of this reduced sample and whether the results on the remaining 33 datasets are representative of the full MulTaBench distribution.

Addressing these points will significantly enhance the statistical rigor and reproducibility of the benchmark's conclusions.

- [writing] The paper reports 95% confidence intervals (CIs) for performance metrics (e.g., Fig 4, Fig 5) but fails to specify the statistical method used for their calculation (e.g., bootstrap, t-distribution, or standard error of the mean). Given the use of 5 random seeds, the method for aggregating variance across seeds and datasets must be explicitly defined to ensure reproducibility.
- [science] The curation pipeline (Sec 3.2) applies a binary acceptance threshold ($\delta=0.001$) to model performance gains. The paper does not report statistical significance tests (e.g., paired t-tests or Wilcoxon signed-rank tests) to verify that these gains are distinguishable from random noise, raising concerns about the robustness of the dataset selection criteria.
- [science] In Appendix A.1, regression targets are discretized into 20 bins for the TAR finetuning step. The paper does not provide a statistical justification for this specific bin count or analyze how this discretization affects the variance and bias of the final regression metrics compared to direct regression finetuning.
- [science] The 'No PCA' ablation (Appendix A.4) excludes datasets with >5 text features and limits analysis to two learners. The paper does not discuss the statistical power of this reduced sample or whether the exclusion criteria introduce selection bias that invalidates the generalizability of the 'no artifact' claim.

paper_reviewer_text_formatting

Verdict: minor_revision

The paper demonstrates strong adherence to LaTeX best practices in most areas, with clear heading hierarchies and well-structured tables. However, several text formatting issues require attention to meet NeurIPS submission standards:

1. **Figure/Caption Consistency**: The Acknowledgments section includes two grant logos (ERC and ELLIOT) embedded via minipages without a figure environment or caption. NeurIPS requires all figures to be numbered and captioned. Either remove these images or convert them to a proper figure environment with a caption (e.g., "Figure X: Funding sources").

2. **Table Formatting**: In Appendix Table 1 (tab:multabench_datasets), the sample size column uses comma-separated numbers (e.g., "1,696"), which may cause rendering inconsistencies in the final PDF. Standard LaTeX tables typically avoid commas in numbers unless explicitly formatted. Consider using a consistent number format (e.g., "1696" or a custom macro for thousands separators).

3. **Symbol Definitions**: Table 1 in Section 3.2 uses $\checkmark$ and $\times$

symbols without a legend in the caption. While the meaning is clear from context, adding a brief note in the caption (e.g., "Checkmarks indicate presence; crosses indicate absence") would improve accessibility and clarity.

4. **Figure Referencing**: The appendix contains wide figures (figure*) that are referenced in the text. Ensure that the figure numbering sequence is correct and that the text references match the actual figure labels. For example, Figure 2 and Figure 3 in the appendix should be properly numbered and referenced as "Figure 2" and "Figure 3" in the text, not as "Figure 1" or "Figure 2" if they are the first figures in the appendix.

5. **Caption Placement**: In Section 5, Figure 4 (fig:leaderboard) is a wide figure with the caption placed below the image. Verify that this placement is consistent with NeurIPS style guidelines, which typically require captions to be placed below figures.

These issues are minor and can be resolved with straightforward edits to the LaTeX source. The overall structure and formatting of the paper are excellent, and these adjustments will ensure full compliance with NeurIPS submission requirements.

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paper_reviewer_writing_quality

Verdict: minor_revision

The paper demonstrates a high level of writing quality, with clear, concise, and well-

structured prose. The introduction effectively sets the stage for the research problem, and the methodology is described with sufficient detail. The flow between sections is generally smooth, and the arguments are logically presented.

However, there are a few minor areas where the writing could be refined for greater precision and academic tone. For instance, in Section 5, the phrase "assuring that the image representations are central" could be improved to "ensuring that the image representations are central" to better convey the intended meaning. Similarly, in Section 6, the sentence "We note that GBDTs exhibit the most substantial gains" is slightly informal; a more formal phrasing like "We observe that GBDTs exhibit the most substantial gains" would be preferable.

In Section 7, the phrase "differing ourselves from existing MMTL benchmarks" is awkward and could be rephrased to "distinguishing our work from existing MMTL benchmarks" or "setting our work apart from existing MMTL benchmarks" for better clarity and flow. Additionally, in the Appendix, under "Text-Tabular Curation," the phrase "datasets which were unavailable due to improper hosting" is somewhat vague; "datasets that were inaccessible due to hosting issues" would be more precise.

Finally, while the term "Target-Aware Representations" (TAR) is consistently capitalized, it is worth ensuring that this capitalization is applied uniformly throughout the paper, especially when the term appears in the middle of sentences or in contexts where lowercase might be more appropriate. These minor adjustments will further enhance the overall readability and professionalism of the manuscript.

- [writing] In Section 5, the phrase 'assuring that the image representations are central' should be corrected to 'ensuring' for better academic tone and precision.
- [writing] In Section 6, the sentence 'We note that GBDTs exhibit the most substantial gains' is slightly informal. Consider rephrasing to 'We observe that GBDTs exhibit the most substantial gains' or 'Our analysis indicates that GBDTs exhibit...'
- [writing] In Section 7, the phrase 'differing ourselves from existing MMTL benchmarks' is awkward. Suggest 'distinguishing our work from existing MMTL benchmarks' or 'setting our work apart from existing MMTL benchmarks'.
- [writing] In the Appendix, Section 'Text-Tabular Curation', the phrase 'datasets which were unavailable due to improper hosting' is slightly vague. Consider 'datasets that were inaccessible due to hosting issues' for clarity.
- [writing] Throughout the paper, ensure consistent capitalization of 'Target-Aware Representations' (TAR). While often capitalized, check if it should be lowercase when not at the start of a sentence or in specific contexts like 'target-aware representations'.